

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 12



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Recap of Previous Lecture



Topic

Pipes and Cistern



Topics to be Covered



Topic-1

Time and Distance



[MCQ]



#Q. A contractor under takes to make a road in 40 days and employs 25 men. After 24 days, he finds that only one-third of the road is made. How many extra men should he employ so that he is able to complete the work 4 days earlier?

- A** 75
- B** 100
- C** 50
- D** None of these

Men	Days	Work
25	24	$\frac{1}{3}$

Assignment

$$100 - 25 = 75$$

$$25 \times \frac{2}{3} \times \frac{1}{1} \times \frac{24}{12} = 100$$

[MCQ]



#Q. A can do a work in 60 days whereas B in 20 days. If they together took a contract of ₹32000, then what would be the share of A?

Assignment

- ☒ A ₹ 8000
- ☐ B ₹ 6000
- ☐ C ₹ 1800
- ☐ D ₹ 2040

$$A : B = 60 : 20 = 3 : 1 \text{ (Days)}$$
$$A's \text{ share} = \frac{1}{4} \times \frac{8000}{32000} = 8000$$
$$A : B = 1 : 3 \text{ (Efficiency)}$$

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

Assignment

- A 4.5 hours
- B 2 hours
- C 6 hours
- D 3.2 hours**

$$A = \frac{1}{4} \quad B = \frac{1}{8} \quad C = -\frac{1}{16}$$

$$A \& B \& C = \frac{1}{4} + \frac{1}{8} - \frac{1}{16} = \frac{4+2-1}{16} = \frac{5}{16}$$

$$\frac{16}{5} = 3\frac{1}{5} \text{ hrs}$$

OR

$$= 3.2 \text{ hrs}$$

[MCQ]



#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

Assignment

$$\text{Total Cap} = 13.5 \times 20$$

$$= \underline{\underline{270 \text{ lit}}}$$

$$\text{No. of Buckets} = \frac{270}{9} = \underline{\underline{30 \text{ Buckets}}}$$

$$20 \times \frac{13.5}{9}$$

Cap	No.
13.5	20
9	?

A

30

B

25

C

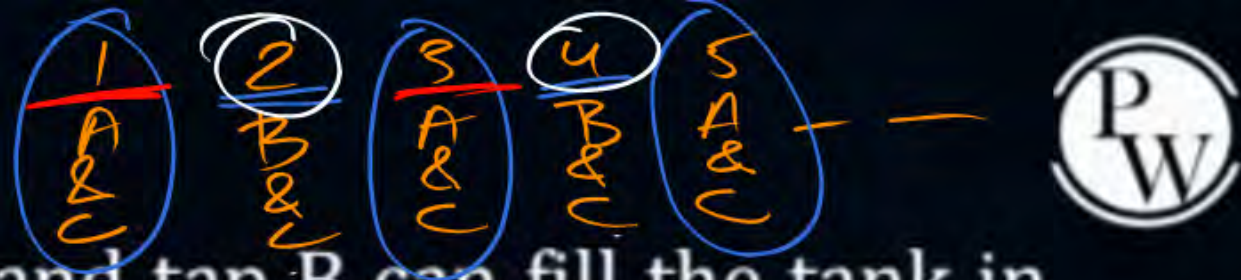
20

D

15

[NAT]

Brainstorming:



#Q. A tank consist of three taps A, B and C. Tap A and tap B can fill the tank in 20hrs and 12 hrs respectively, whereas tap C can empty the tank in 18hrs. C was kept open whereas A and B are opened in alternate hours i.e. 1st hr A, 2nd hr B, 3rd hr A.....so on. In how many hours the tank would be filled?

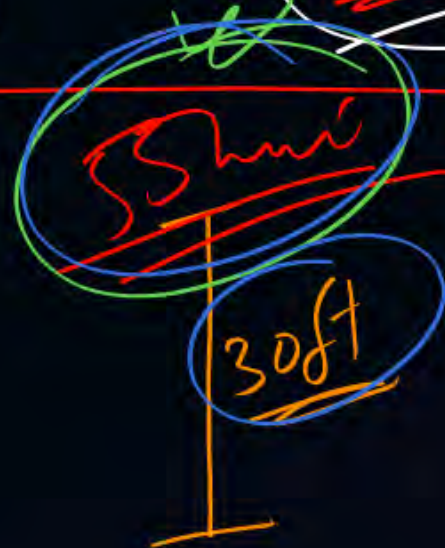
$$A = \frac{1}{20} \quad | \quad B = \frac{1}{12} \quad | \quad C = \frac{1}{18}$$

1st loop A & C $\rightarrow \frac{1}{20} - \frac{1}{18} = \frac{-1}{180}$

2nd loop B & C $\rightarrow \frac{1}{12} - \frac{1}{18} = \frac{5}{180} = \frac{1}{36}$

1 min $\rightarrow$ $\frac{+384}{-284}$ } $+184$
 2 min $\rightarrow$ $\frac{+384}{-284}$ }
2nd & 3rd

$89\frac{4}{5}$ hrs



$$\frac{5}{180} - \frac{1}{180} = \frac{4}{180} = \frac{1}{45} \rightarrow \underline{\underline{2 \text{ hrs}}}$$

$\frac{1}{45}$

2 min $\rightarrow$ 1 ft

54 min $\rightarrow$ 27 ft

56 min $\rightarrow$ 28 ft

+3

BCC can do 1 work

$\frac{44}{45} \rightarrow 88 \text{ hrs}$
 $+15$ 1st hr

$\frac{44}{45} \rightarrow$

89 hrs

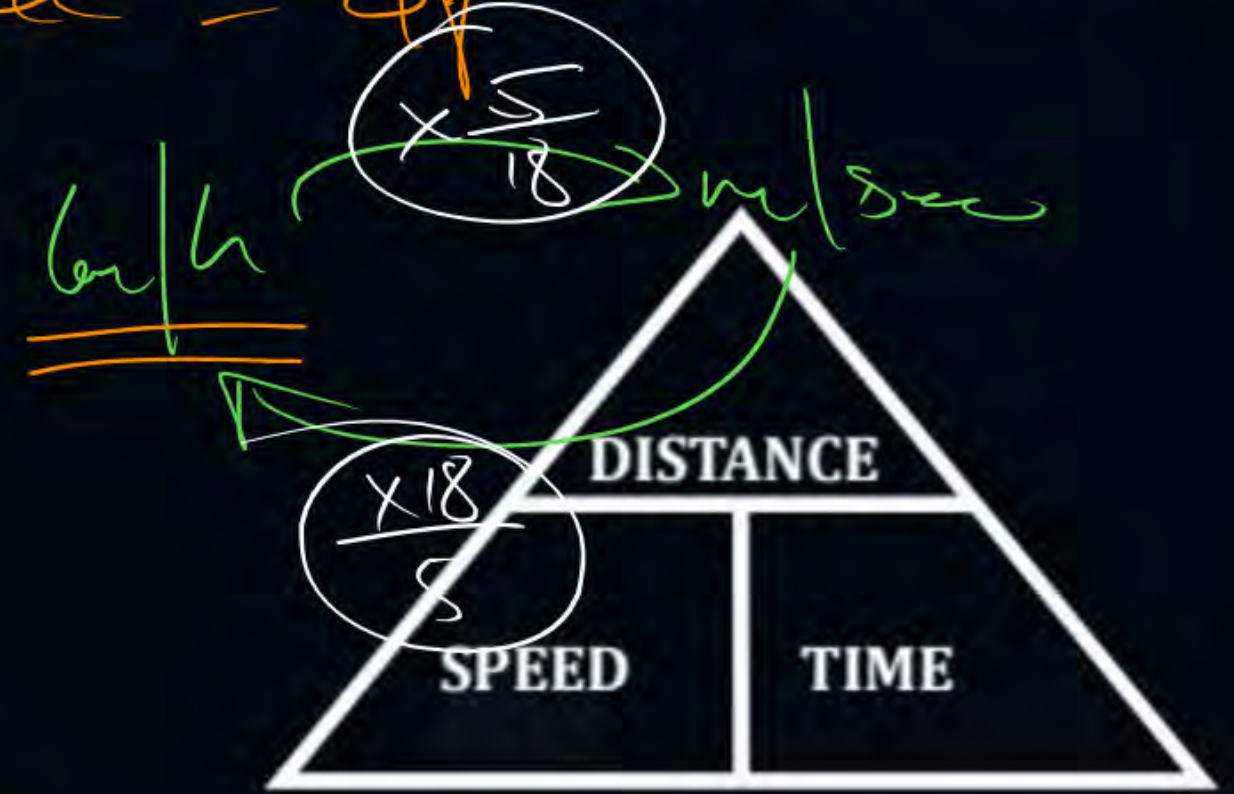
$= 36 \text{ hrs}$
 $\frac{1}{45} \times 88 = \frac{88}{45} = \frac{4}{5} \text{ hrs}$

Time & Distance



$$(D) = (S) \times (T)$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$



$$T = \frac{D}{S}$$

$$S = \frac{D}{T}$$

I $\Rightarrow$ Average Speed

II $\Rightarrow$ Relative Speed

$$\frac{5}{18} \frac{1000 \text{ m}}{3600 \text{ sec}}$$

$$D = 800 \text{ m}$$

$$S = 40 \text{ km/hr}$$

$$T = \frac{800}{40} = 20 \text{ m/hr/hr?}$$

Q If the T.G. + 1 & H.H. of a clock

conceides at an every interval of 64 minutes
 then how much does the clock lose or
 u Normal
 gain in a day

O.P. Council

12 hrs $\rightarrow$ 11 times

24 hrs $\rightarrow$ 22 times

$$\frac{360}{5.5} = \frac{720}{11} = \underline{\underline{65 \frac{5}{11} \text{ minutes}}}$$



$$\frac{16}{11} \times \underline{\underline{22^2}}$$

$$\underline{\underline{+32 \text{ min}}}$$

$$(+1) \frac{5}{11} \text{ min} \underline{\underline{\frac{(+16)}{11}}}$$

[MCQ]

$$T = \frac{D}{S}$$



#Q. If you travel from M to N at 40km/hr and N to M at 60 km/hr. What would be your average speed of the journey?



$$\frac{x}{40} + \frac{x}{60} = \frac{2x}{\text{A.S.}}$$

$$\Rightarrow \frac{\cancel{8x}}{\cancel{120}} = \frac{\cancel{2x}}{\text{A.S.}}$$

24

$$\frac{2xy}{x+y}$$

$$\frac{2 \times 40 \times 60}{100} = 48 \text{ kmph}$$

$$\text{A.S.} = 24 \times 2 = \underline{\underline{48 \text{ kmph}}}$$

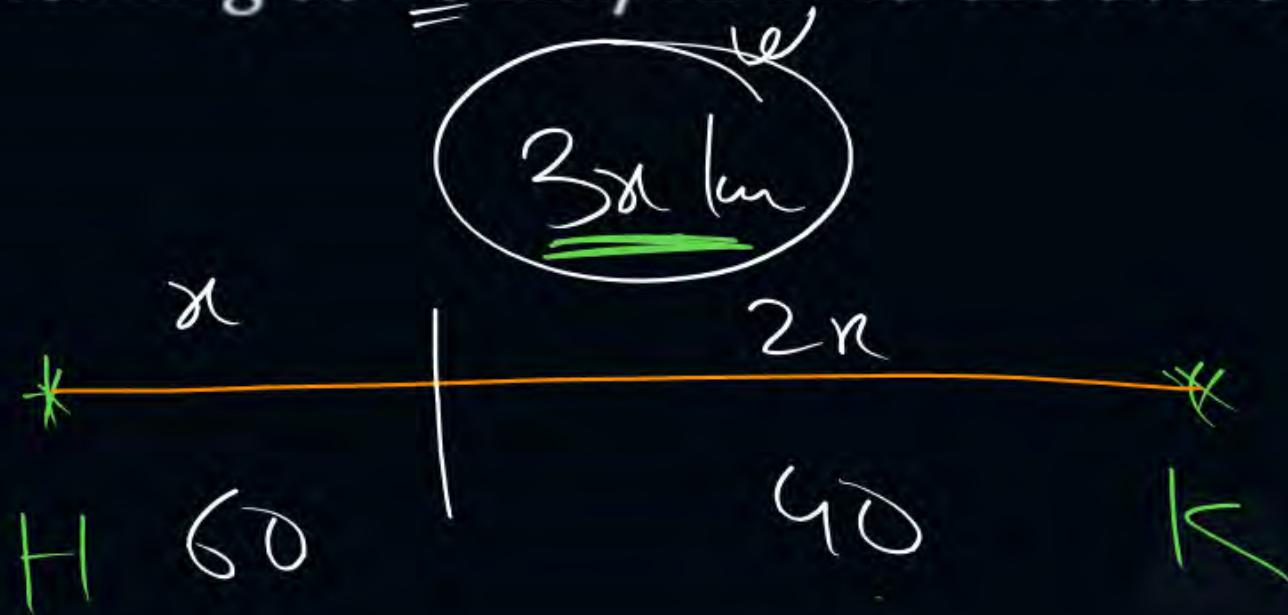
[MCQ]

$$A.S. = \frac{\text{Total Dist. (T.D.)}}{\text{Total Time (T.T.)}}$$

$$T.T. = \frac{T.D.}{A.S.}$$



#Q. A man travels from Howrah to Kanpur in his car. If he covers $\frac{1}{3}$ rd of the distance at 60 km/hr & remaining at 40 km/hr. Find the average speed of his journey.



$$\frac{x}{60} + \frac{2x}{40} = \frac{3x}{A.S.}$$

$$\frac{2x + 6x}{120} = \frac{3x}{A.S.} \Rightarrow \frac{8x}{120} = \frac{3x}{A.S.}$$

$$A.S. = 15 \times 3 = 45 \text{ km/hr}$$

[MCQ]



#Q. A man travels $\frac{1}{4}$ th of his journey at 20 km/hr, another $\frac{1}{4}$ th at 30 km/hr & remaining at 60 km/hr. Find the average speed of his journey.

Diagram illustrating the journey segments:

$$\frac{x}{20} + \frac{x}{30} + \frac{2x}{60} = \frac{4x}{A.S.}$$

$$\Rightarrow \frac{7x}{60} = \frac{4x}{A.S.}$$

$$\therefore A.S. = \frac{60 \times 4}{7} = \frac{240}{7} = 34\frac{2}{7} \text{ OR } 34.28$$

[MCQ]

#Q. If a car travels along four sides of a square at 100 kmph, 200 kmph, 300 kmph and 400 kmph. Find its average speed.

Assignment

- A** 152 km/hr
- B** 180 km/hr
- C** 192 km/hr
- D** 210 km/hr

[MCQ]

#Q. A tourist covers half of his journey by train at 60 km/hr, half of the remainder by bus at 30 km/hr and the rest by cycle at 10 km/hr. Average speed of the tourist during the journey is

Assignment

- A** 36 km/hr
- B** 33 km/hr
- C** 24 km/hr
- D** 18 km/hr

[MCQ]



#Q. A person covers a certain distance at a certain speed. If he decrease his speed by 20%, then he takes 16 minutes more to cover the same distance. Find the time taken by him to cover the original distance at original speed.

$$\begin{aligned} \frac{\cancel{100}^5}{\cancel{80}_4} T - T &= 16 \\ \left(\frac{5}{4} - 1 \right) T &= 16 \\ \frac{1}{4} T &= 16 \end{aligned} \quad \bigg| \quad T = 16 \times 4 = \boxed{64 \text{ min}}$$

$D = S \times T$

$\frac{2}{4} \quad \frac{1}{2}$

$\frac{80}{100} \quad \frac{100}{80}$

[MCQ]



#Q. Walking $\frac{3}{4}$ th of the usual speed a man is late to office by 10 minutes, find the usual time.

Assignment

- A** 40 minutes
- B** 12 minutes
- C** 60 minutes
- D** 30 minutes

[MCQ]

#Q. Mumbai police observes a thief 800 metres away from him. The thief started running at 80 km/hr & the police at 170 km/hr. In how much time the police can catch the thief?

Assignment

- A** 8.88 minutes
- B** 8.88 hours
- C** 32 seconds
- D** 32 minutes

[MCQ]

#Q. An employee goes to his office from his house at a speed of 60 km/hr and reaches his office 10 minutes late. If he follows the speed of 80 km/hr, he reaches his office 5 minutes early. Find the distance he covers to reach his office.

Assignment

- A** 55 km
- B** 45 km
- C** 50 km
- D** 60 km



2 mins Summary



Topic

Time & Distance

(Basic)

Average Speed

Relative Speed

THANK - YOU